Table 1: Continuous vs traditional profiling

Continuous profiling	Traditional profiling
Runs continuously	Performed manually
Long-term monitoring	Short-duration analysis
Production-focused	Development-focused
Historical trends available	Snapshot-based
Automated collection	High intervention

- Improve application responsiveness

Table 1 describes the difference between traditional and continuous profiling.

Architecture of Grafana Pyroscope

Grafana Pyroscope follows a distributed continuous profiling architecture designed to collect, process, store, query, and visualise profiling data from applications running in production environments. The architecture of the tool consists of five main layers (Figure 1).

Applications and Linux infrastructure is the lowest layer of the architecture, where business applications like Java, Python, Go Services, Node.js, .NET, Kubernetes Pods, and Linux servers are executed. A typical workload in this layer includes web applications, microservices, databases, APIs, background jobs, and cloud services, all of which consume resources like CPU, memory, disk I/O, network I/O, threads, and locks. Pyroscope continuously observes these resources.

The second layer is the **profiling agent layer,** which collects execution data from running applications. This layer is responsible for various tasks such as:

- Sampling stack traces
- Monitoring memory allocations
- Tracking CPU usage
- Collecting profiling events
- Sending data to the Pyroscope server

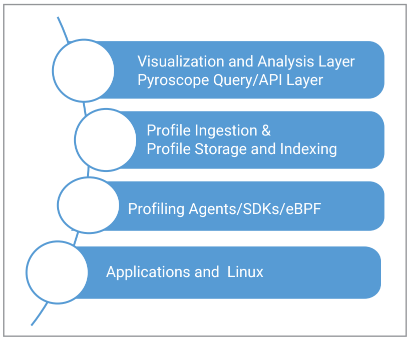

Figure 1: Layers of Grafana Pyroscope

It comprises three components: language-specific SDKs, eBPF profilers, and a sampling engine. Language-specific SDKs are provided for multiple languages. For example:

```
import pyroscope
pyroscope.configure(application_
name="ecommerce-app",

server_address=http://pyroscope-
server:4040
)
```

Modern Pyroscope deployments can utilise Linux eBPF technology. And instead of tracing every function call, Pyroscope uses statistical sampling.

The **profile storage and indexing layer** is the core component of Grafana Pyroscope responsible for storing, organising, compressing, indexing, and retrieving profiling data collected from applications. Without an optimised storage and indexing mechanism, continuous profiling data would quickly consume excessive storage space and become difficult to analyse.

The **profile ingestion and processing layer** is the main part of Grafana Pyroscope and connects the profiling agents to the storage layer. Its main job is to get profile information from apps, check if it's correct, work with the data, group similar profiles together, make the data smaller, add extra details to it, and get it ready to be stored and used quickly. In a big production setting, thousands of applications can keep sending profiling data every second. This layer makes sure this data is managed smoothly without overloading the storage system.

The **Pyroscope query/API layer** is the component that sits between the storage layer and the visualisation layer. Retrieving, processing, aggregating, and delivering profiling data required by users, Grafana dashboards, APIs, and other observability tools is its main duty. The Query/API layer is in charge of effectively receiving the profiling data and providing insightful performance information, while the ingestion layer is in charge of storing the data. Serving as the 'brain' of the retrieval system, it transforms user requests into optimal storage searches and provides data in formats appropriate for call graphs, differential profiles, flame graphs, and historical analysis.

The highest tier of the Grafana Pyroscope architecture is called the **visualisation and analysis** tier. It converts unprocessed profiling data into insightful visuals and useful information. The visualisation and analysis layer helps developers, system administrators, DevOps engineers, and SREs understand application behaviour, find performance bottlenecks, and optimise system performance, while the lower layers gather, process, store, and retrieve profile data. This layer is the main interface between users and profiling data. It offers timelines, differential views, flame graphs, call graphs, dashboards, and analytical tools that transform complicated profile data into understandable representations.

Advantages of Grafana Pyroscope

One of the most significant advantages of Pyroscope is continuous profiling. Other benefits are:

- Profiles applications 24×7.
- Captures historical performance data.
- Enables analysis of intermittent issues.